26/02/2021

Code-A

Aakash
Medical | IIT-JEE | Foundations

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MM : 720 **NEET BOOSTER TEST SERIES (NBTS) for NEET-2021** Time : 3.00 Hrs.

Test - II

Topics Covered : Complete Syllabus of Class-XI & XII (NEET Pattern)

Instructions :

(i) Use blue/black ballpoint pen only to darken the appropriate circle.

(ii) Mark should be dark and should completely fill the circle.

(iii) Dark only one circle for each entry.

(iv) Dark the circle in the space provided only.

(v) Rough work must not be done on the Answer sheet and do not use **white-fluid** or any other **rubbing material** on Answer sheet.

(vi) Each question carries 4 marks. For every wrong response 1 mark shall be deducted from total score.

PHYSICS

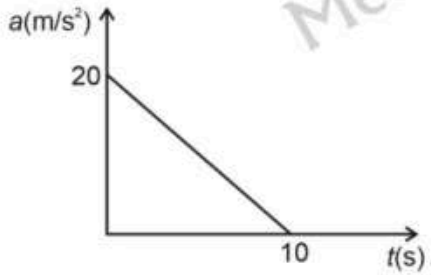
Choose the correct answer :

1. Electron volt is the unit of

(1) Power (2) Potential

(3) Electric charge (4) Energy

2. A particle starts from rest. If acceleration (a) versus time (t) graph is as shown in the figure, then the maximum speed of the particle will be



(1) 50 m/s (2) 100 m/s

(3) 60 m/s (4) 120 m/s

3. A body is projected with velocity 40 m/s making an angle 30° with the horizontal. The vertical component of its velocity after 3 s is (g = 10 m/s²)

(1) 10 m/s downward (2) 10 m/s upward

(3) 5 m/s downward (4) 5 m/s upward

4. The tension in the cable of 800 kg elevator is 8000 N. The elevator (g = 10 m/s²)

(1) Is ascending upwards

(2) Is descending downwards

(3) May be at rest or accelerating

(4) May be at rest or in uniform motion

5. A particle of mass m_1 moving with velocity v strikes elastically with a mass m_2 at rest, then the condition for maximum transfer of kinetic energy is

(1) $m_1 >> m_2$ (2) $m_2 >> m_1$

(3) $m_1 = m_2$ (4) $m_1 = 2m_2$

6. When a mass 'm' is attached to a vertical spring and slowly released, the extension in the spring is 'x' when a mass '$3m$' is attached to the same spring and released suddenly, the maximum extension in the spring is

(1) $3x$ (2) $6x$

(3) $2x$ (4) $4x$

7. If the linear density of a rod of length L varies as $\lambda = kx$ where k is a constant and x is the distance of any point from one end placed at origin. The distance of centre of mass from $x = 0$ is

(1) $\frac{2L}{3}$ (2) $\frac{L}{3}$

(3) $\frac{L}{4}$ (4) $\frac{3L}{4}$

8. A uniform cylinder is released from the top of an incline of inclination 30° and length 'ℓ'. If the cylinder rotate without slipping, its speed at the bottom is

(1) $\sqrt{\frac{2g\ell}{3}}$ (2) $\sqrt{\frac{3g\ell}{4}}$

(3) $\sqrt{\frac{8g\ell}{3}}$ (4) $\sqrt{\frac{2g}{3\ell}}$

9. A loop is of radius 2 m and mass 50 kg. It rolls along a horizontal floor so that its centre of mass has speed 10 cm/s. How much work has to be done to stop it.

(1) 1 J (2) 4 J

(3) 0.5 J (4) 8 J

10. The displacement equation of SHM is $y = 10\sin\left(2\pi t + \frac{\pi}{3}\right)$m. The initial speed of the particle is

(1) 20π m/s (2) 10π m/s

(3) 5 m/s (4) 10 m/s

11. A spherical planet in space has mass M_0 and diameter D_0. A particle of mass 'm' falling freely near the surface of this planet will experience an acceleration due to gravity which is equal to

(1) $\frac{GM_0}{D_0^2}$ (2) $\frac{GmM_0}{D_0^2}$

(3) $\frac{4GmM_0}{D_0^2}$ (4) $\frac{4GM_0}{D_0^2}$

12. A metal wire of density d floats on water. The maximum radius of the wire so that it may not sink is (T is surface tension of water)

(1) $\sqrt{2\pi dgT}$ (2) $\sqrt{\frac{2T}{\pi dg}}$

(3) $\sqrt{\frac{\pi dg}{2T}}$ (4) $\sqrt{\frac{2Tg}{\pi d}}$

13. When a bimetallic strip made of iron (α_1) and copper α_2 ($\alpha_2 > \alpha_1$) is heated

(1) Its length does not change

(2) It gets twisted

(3) It bends with iron on concave side

(4) It bends with iron on convex side

14. A graph between pressure P and $\frac{1}{V}$ (where V is volume) for an ideal gas at constant temperature is

(1) Parabola

(2) Rectangular hyperbola

(3) Ellipse

(4) Straight line

15. The molar heat capacity in process of diatomic gas if it does a work $\frac{Q}{3}$, when a heat Q is supplied to it, is ($\gamma = 7/5$)

(1) $\frac{15R}{4}$ (2) $\frac{15R}{2}$

(3) $\frac{9R}{4}$ (4) $\frac{9R}{2}$

16. The power radiated by a black body is P and it radiates maximum energy around the wavelength λ_0. If temperature of the black body is changed so that it radiate maximum energy around a wavelength $\frac{2\lambda_0}{3}$, the power radiated by it will increase by a factor of

(1) $\frac{27}{8}$ (2) $\frac{64}{27}$

(3) $\frac{81}{16}$ (4) $\frac{256}{81}$

17. Equal temperature difference exists between the end of two metallic rods M_1 and M_2 of equal length. Their thermal conductivities $2K$ and K and cross-sectional areas are respectively A_1 and A_2. If rate of heat transfer is equal for both rod then value of $\frac{A_1}{A_2}$ will be

(1) $\frac{1}{2}$ (2) 1

(3) $\frac{1}{4}$ (4) $\frac{1}{3}$

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18. A wave of wavelength 1 m is superposed on reflected wave to form a stationary wave. A node is located at x = 2 m. The next node will be located at x is equal to

(1) 3 m (2) 3.5 m

(3) 4 m (4) 2.5 m

19. The first overtone of an open pipe has frequency n. The third harmonic of a closed pipe of same length will have frequency

(1) $\frac{3n}{4}$ (2) $\frac{2n}{4}$

(3) $\frac{3n}{2}$ (4) $\frac{n}{3}$

20. A stretched string 0.5 m long has a third harmonic frequency 900 Hz. The velocity of the wave in stretched strings is

(1) 150 m/s (2) 200 m/s

(3) 300 m/s (4) 100 m/s

21. A ray of light passes normally through a slab (μ = 5/3) of thickness 't'. If the speed of light in vacuum be 'c' then time taken by the ray to cross the slab will be

(1) $\frac{2t}{5c}$ (2) $\frac{3t}{5c}$

(3) $\frac{4t}{5c}$ (4) $\frac{5t}{3c}$

22. A concave mirror gives an image two times as larger as the object placed 15 cm from it. For the image to be real, the focal length should be

(1) 15 cm (2) 30 cm

(3) 20 cm (4) 10 cm

23. A person can see things clearly upto 2 m. What is the power of lens he should use so that he can see upto 10 m?

(1) –1.5 D (2) –2.5 D

(3) +1.5 D (4) +2.5 D

24. When a thin metal plate is placed in path of one of the interfering beams of light

(1) The fringes becomes blurred

(2) The fringes becomes brighter

(3) The fringes will disappear

(4) The fringe width increases

25. Two polarizers have their axes inclined at 45° to each other. If unpolarised light of intensity I_0 is incident on the first polarizer, then the intensity transmitted through second polarizer is

(1) I_0 (2) $\frac{I_0}{4}$

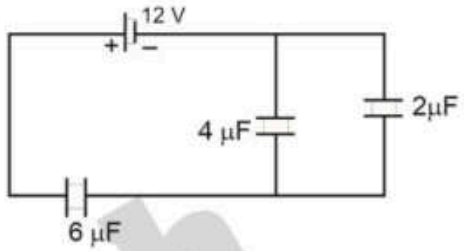
(3) $\frac{I_0}{2}$ (4) Zero

26. The electric potential V is given as function of distance x (metre) by $V = (4x^2 + 3x - 2)$ volt. The magnitude of force on +2 C charge placed at x = 2 m will be

(1) 13 N (2) 38 N

(3) 11 N (4) 22 N

27. The energy stored on 4 μF capacitor in the circuit shown is



(1) 36 μJ (2) 24 μJ

(3) 48 μJ (4) 72 μJ

28. With a potentiometer null points were obtain at 120 cm and 140 cm with cell of emf 2.4 V and one unknown emf x volt. Unknown emf is

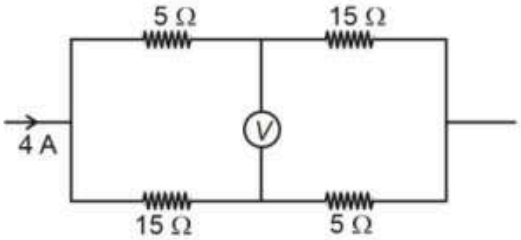
(1) 2.6 V

(2) 2.8 V

(3) 1.4 V

(4) 1.8 V

29. In the circuit as shown in figure, the reading of the volt meter V is



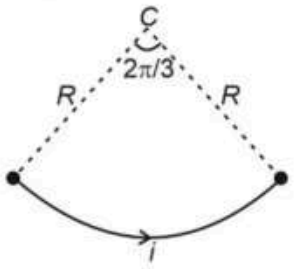
(1) 10 V

(2) 15 V

(3) 5 V

(4) 20 V

30. A current i ampere flows in a circular arc of wire whose radius is R, which subtends an angle $\frac{2\pi}{3}$ radian at its centre. The magnetic induction B at the centre is



(1) $\frac{\mu_0 i}{3R}$ (2) $\frac{\mu_0 i}{6R}$

(3) $\frac{\mu_0 i}{4R}$ (4) $\frac{\mu_0 i}{2R}$

31. If two bar magnets of different magnetic lengths have equal magnetic moments, then the pole strength is

(1) Equal for both the magnets

(2) Lesser for shorter magnet

(3) More for longer magnet

(4) More for shorter magnet

32. The period of oscillation of a magnet at a place is 4 s. When it is remagnetised, so that the pole strength becomes four times the initial value, the period of oscillation in seconds is

(1) $\frac{1}{2}$ (2) 1

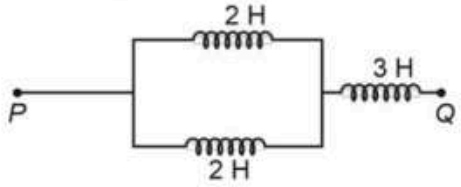
(3) 2 (4) 3

33. The magnetic flux of 500 μWb per turn passing through a 200 turn coil reversed in 20×10^{-3} s. The average emf induced in the coil in volt is

(1) 2.5 (2) 7.5

(3) 5.0 (4) 10.0

34. The equivalent inductance between points P and Q in figure is



(1) 2 H (2) 1 H

(3) 4 H (4) 3 H

35. A L-C circuit has natural frequency f. If capacitance and inductance both are halved, then the natural frequency would become

(1) $\frac{f}{2}$ (2) $2f$

(3) $4f$ (4) $\frac{f}{4}$

36. The wavelength of the matter wave (de-Broglie) does not depend on

(1) Momentum (2) Velocity

(3) Mass (4) Charge

37. In photoelectric phenomenon, at a given frequency ($\nu > \nu_0$), the number of photoelectrons emitted depends on

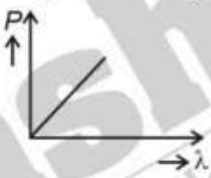
(1) The intensity of incident radiation

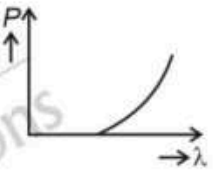
(2) The velocity of incident radiation

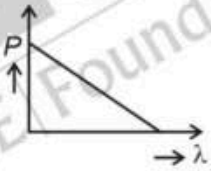
(3) The work function of photocathode

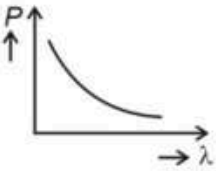
(4) Both (2) and (3)

38. Which of the following figures represent the variation of particle momentum(P) and associated de-Broglie wavelength(λ)?

(1) 

(2) 

(3) 

(4) 

39. The radius of third stationary orbit in Bohr's atom model is R. The radius of fourth orbit in the same atom will be

(1) $\frac{16R}{9}$ (2) $\frac{4R}{3}$

(3) $\frac{3R}{2}$ (4) $\frac{3R}{4}$

40. If R is the Rydberg constant for hydrogen atom, then the wave number of H_β line of Lyman's series is

(1) $\frac{3R}{4}$ (2) $\frac{8R}{9}$

(3) $\frac{R}{4}$ (4) $\frac{3R}{2}$

41. Activity of radioactive element decreases to one third of original activity I_0 in 3 years. After further 9 year its activity will be

(1) $\frac{I_0}{9}$ (2) $\frac{I_0}{18}$

(3) $\frac{I_0}{27}$ (4) $\frac{I_0}{6}$

42. What is the approximate energy released by the complete annihilation of an alpha particle?

(1) 3724 MeV (2) 372 J

(3) 931 MeV (4) 931 J

43. At absolute zero temperature, a crystal of pure Germanium

(1) Behaves as a perfect conductor

(2) Behaves as a semiconductor

(3) Behaves as a perfect insulator

(4) Behaves as a superconductor

44. For a transistor $x = \frac{1}{\alpha}$ and $y = \frac{1}{\beta}$ where α and β are current gains in common base and common emitter configuration. Then

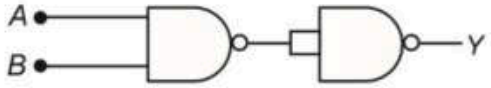
(1) $x + y = 1$

(2) $x - y = 1$

(3) $2x = 1 - y$

(4) $x + y = 0$

45. The arrangement shown in figure performs the logic function of



(1) AND gate

(2) NAND gate

(3) OR gate

(4) XOR gate

CHEMISTRY

46. Paramagnetic species having bond order of 2.5 is

(1) C_2^{2-} (2) O_2^-

(3) O_2^+ (4) O_2

47. In the Kjeldahl's method for estimation of nitrogen present in an organic compound, ammonia evolved from 0.5 g sample, neutralized 20 ml of 0.5 M H_2SO_4. The percentage of nitrogen in the organic compound is

(1) 28% (2) 56%

(3) 46% (4) 92%

48. Which one of the following is most reactive towards electrophilic attack?

(1) 

(2) 

(3) 

(4) 

49. Non-aromatic species among the following is

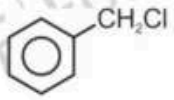
(1) 

(2) 

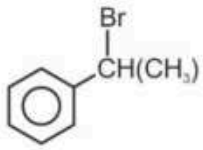
(3) 

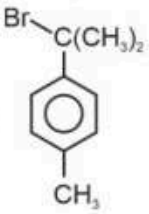
(4) 

50. Which of the following compounds will undergo racemisation in presence of aqueous alkali?

(1) 

(2) 

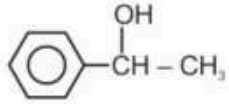
(3) 

(4) 

51. Alcohol which will react at slowest rate with $ZnCl_2$ and Conc. HCl is

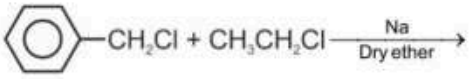
(1) CH_3CH_2OH

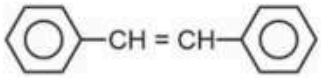
(2) $(CH_3)_3C - OH$

(3) 

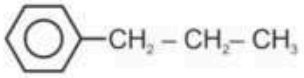
(4) $(CH_3)_2CH - OH$

52. Which is observed as one of the products in given reaction?



(1) 

(2) $CH_3CH_2CH_3$

(3) 

(4) $CH_3CH_2CH_2CH = CH_2$

53. Select the incorrect statement among the following

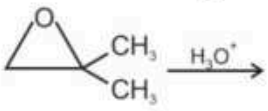
(1) Among halogens chlorine has highest electron affinity

(2) $2p_x$ orbital has one nodal plane

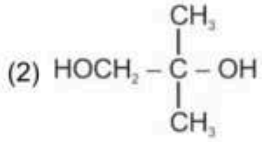
(3) Li has highest first ionization enthalpy among alkali metals

(4) $3d_{x^2-y^2}$ has three radial nodes

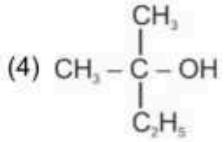
54. Major product of the following reaction is



(1) CH_3CH_2CHO

(2) 

(3) $HO - CH_2 - CH_2OH$

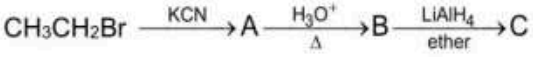
(4) 

55. Glucose does not react with

(1) $NaHSO_3$ (2) $(CH_3CO)_2O$

(3) NH_2OH (4) Br_2 water

56. In the following sequence of reactions



The major end product (C) is

(1) Ethyl alcohol

(2) Propyl alcohol

(3) Isopropyl alcohol

(4) Butyl alcohol

57. Most acidic compound among the following is

(1) HCOOH (2) C_6H_5COOH

(3) CH_3COOH (4) NO_2CH_2COOH

58. The pair of compounds which cannot be distinguished by haloform test is

(1) CH_3CHO and $CH_3CH_2CH_2OH$

(2) CH_3OH and HCHO

(3) CH_3CH_2CHO and $CH_3COCH_2CH_2CH_3$

(4) CH_3CHO and $CH_3CH_2COCH_2CH_3$

59. Spin only magnetic moment of $[CoF_6]^{3-}$ ion in BM is

(1) Zero (2) $\sqrt{35}$

(3) $\sqrt{24}$ (4) $\sqrt{15}$

60. Which of the following will not react with sodium hydrogen carbonate?

(1) Benzoic acid

(2) 2, 4-dinitrophenol

(3) Phenol

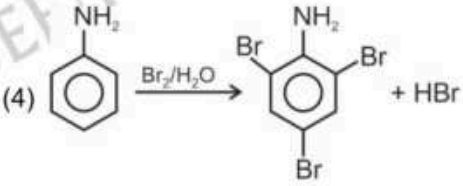
(4) 4-nitrobenzoic acid

61. Carbylamine reaction among the following is

(1) $CH_3CH_2NH_2 \xrightarrow[(ii)H_2O]{(i)\ NaNO_2+HCl} CH_3CH_2OH + N_2\uparrow + HCl$

(2) $CH_3CH_2NH_2 + CHCl_3 + KOH \xrightarrow{\Delta} CH_3CH_2NC + KCl + H_2O$

(3) $CH_3NH_2 + C_6H_5COCl \rightarrow CH_3NHCOC_6H_5 + HCl$

(4) 

62. Select the amine which does not react with benzenesulphonyl chloride?

(1) CH_3NH_2

(2) $(CH_3)_2NH$

(3) $Ph - NH - CH_3$

(4) $(CH_3)_3N$

63. Percentage strength of H_2O_2 solution having volume strength 5.6 is

(1) 1.7 (2) 2.4

(3) 17 (4) 24

64. By what factor does the root mean square speed of molecules increase when temperature (in Kelvin) is doubled while molar mass of gas becomes half?

(1) 1.414 (2) 2

(3) 1.732 (4) 2.828

65. If the dissociation energy of $CH_4(g)$ is 360 kcal mol^{-1} and that of $C_3H_8(g)$ is 880 kcal mol^{-1}, then the 'C – C ' bond energy (in kcal mol^{-1}) is

(1) 40 (2) 80

(3) 120 (4) 90

66. 2 mol of an ideal gas expanded reversibly from 2 L to 200 L at 300 K. Work done in this process is

(1) – 2763.6 R (2) – 1381.8 R

(3) – 5527.2 R (4) – 4000 R

67. Which among the following is a common component of photochemical smog?

(1) Formaldehyde (2) Methanol

(3) SO_2 (4) CO_2

68. Which pair of solutions can act as acidic buffer?

(1) HCOOH and HCl

(2) Na_2SO_4 and H_2SO_4

(3) H_3PO_4 and NaH_2PO_4

(4) Na_3PO_4 and NaOH

69. Salt which undergoes only anionic hydrolysis is

(1) NaCl (2) CH_3COONH_4

(3) CH_3COONa (4) NH_4Cl

70. Solubility of SrF_2 ($K_{sp} = 2.5 \times 10^{-9}$) in the aqueous solution of 0.1 M KF is

(1) 2.5×10^{-6} M (2) 2.5×10^{-7} M

(3) 1.25×10^{-7} M (4) 1.25×10^{-6} M

71. pH of solution which is formed by mixing equal volume of 0.1 M H_2SO_4 and 0.1 M NaOH solutions at 25°C is (log 5 = 0.7)

(1) 7 (2) 1.3

(3) 2.3 (4) 1.7

72. Average oxidation states of sulphur in $H_2S_2O_3$, $H_2S_2O_6$ and $H_2S_2O_8$ respectively are

(1) +2, +5 and +7 (2) +2, +2 and +6

(3) +2, +4 and +6 (4) +2, +5 and +6

73. Standard reduction potentials of the half reactions are given below

$Ag^+ + e^- \longrightarrow Ag$	$E^o = 0.8$ V
$Cr^{3+} + 3e^- \longrightarrow Cr$	$E^o = -0.74$ V
$Zn^{2+} + 2e^- \longrightarrow Zn$	$E^o = -0.76$ V

The strongest oxidising and reducing species respectively are

(1) Ag^+, Zn^{2+} (2) Ag, Cr

(3) Ag, Cr^{3+} (4) Ag^+, Zn

74. Orbital angular momentum of $4d_{x^2-y^2}$ orbital is

(1) $\sqrt{6}\frac{h}{2\pi}$ (2) $\frac{h}{2\pi}$

(3) $\sqrt{3}\frac{h}{2\pi}$ (4) $\frac{h}{\pi}$

75. Number of Cl^- in a CsCl unit cell is

(1) 1 (2) 2

(3) 4 (4) 8

76. Mole of $CO_2(g)$ present in 900 ml of soda water when packed under 1.6 atm pressure at 25°C is (K_H for $CO_2(g)$ at 25°C is 1.6×10^3 atm)

(1) 0.01 mol (2) 2 mol

(3) 0.05 mol (4) 0.5 mol

77. Freezing point of an aqueous solution of 0.1 m Na_2SO_4 is (K_f(water) = 1.86 K $molal^{-1}$; consider complete dissociation of electrolyte)

(1) 270.2 K (2) 272.4 K

(3) 273.6 K (4) 269.2 K

78. The vapour pressure of two liquids A and B respectively are 1000 and 800 torr. Ratio of mole fractions of A and B in vapour mixture is, if liquids A and B are added in the mole ratio of 2 : 5

(1) 2 : 3 (2) 2 : 5

(3) 1 : 8 (4) 1 : 2

79. If 80% of a first order reaction was completed in 80 minutes then 50% of the same reaction would be completed in (log 5 = 0.7, log 3 = 0.48)

(1) 40 minutes (2) 70 minutes

(3) 64.5 minutes (4) 34.4 minutes

80. E_{cell} of cell, Ag/Ag^+ (0.01 M) || Ag^+(0.1 M)| Ag, at 298 K is ($E^o_{Ag^+/Ag}$ = 0.8 V)

(1) 0.0591 V (2) 0.1182 V

(3) – 0.0591 V (4) – 0.1182 V

81. On electrolysis of highly concentrated solution of $CuSO_4$ (using platinum electrodes), cathode and anode products respectively are

(1) Cu(s), SO_2(g) (2) Cu(s),$S_2O_8^{2-}$(aq)

(3) H_2(g), O_2(g) (4) Cu(s), O_2(g)

82. Ion of highest coagulation power for the coagulation of TiO_2 sol is

(1) Al^{3+} (2) Ba^{2+}

(3) SO_4^{2-} (4) $[Fe(CN)_6]^{4-}$

83. Select the incorrect statement among the following

(1) Blister appearance of copper is due to evolution of SO_2(g) from it

(2) Copper matte contains Cu_2O and FeS

(3) In distillation (refining process) generally pure metal is obtained in distillate

(4) In leaching of silver, zinc metal acts as a reducing agent

84. Number of P–H and P–OH bond(s) present respectively in hypophosphoric acid is

(1) 1, 4 (2) 2, 4

(3) 0, 4 (4) 2, 3

85. Select the incorrect order of property mentioned for hydrogen halides among the following

(1) HF < HCl < HBr < HI (Bond length)

(2) HF < HCl < HBr < HI (Acidic strength)

(3) HF > HI > HBr > HCl (Boiling point)

(4) HF > HI > HBr > HCl (Melting point)

86. Partial hydrolysis of XeF_6 produces

(1) XeO_3 (2) XeF_4

(3) $XeOF_4$ (4) XeO_4

87. Which element has least density?

(1) Cr (2) Mn

(3) Zn (4) Sc

88. Most basic hydroxides among the following is

(1) $Eu(OH)_3$ (2) $Gd(OH)_3$

(3) $Ce(OH)_3$ (4) $Tm(OH)_3$

89. Optical isomerism is shown by

(1) trans-$[Pt(en)_2Cl_2]$

(2) fac-$[Co(NH_3)_3Br_3]$

(3) cis-$[Co(en)_2Cl\ Br]I$

(4) $K_2[NiCl_6]$

90. Crystal field stabilising energy (CFSE) for high spin octahedral complex containing metal of d^5 configuration is

(Δ_0 : Octahedral splitting energy gap, P : pairing energy per pair of electrons)

(1) – 0.4 Δ_0 (2) Zero

(3) – 2.4 Δ_0 (4) – 2.4 Δ_0 + 3P

BOTANY

91. Choose the **incorrect** statement from the following

(1) Unicellular as well as multicellular organisms grow by cell division

(2) Reproduction is essential for the survival of an individual

(3) Metabolism is not exhibited in non-living objects

(4) Living organisms are aware of their surroundings

92. Almost all organisms which have organ level of body organisation are

(1) Only heterotrophic

(2) Only autotrophic

(3) Either autotrophic or heterotrophic

(4) Chemosynthetic as well as saprophytic

93. All of the following taxon can be for the same plant, **except**

(1) Monocotyledonae

(2) Poales

(3) Solanaceae

(4) Angiospermae

94. Select the **incorrect** match from the following

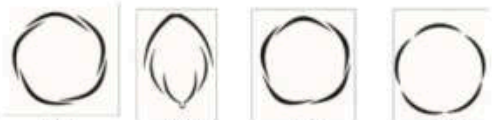
(1)	Euglenoids	–	Have pigments identical to that of higher plants
(2)	Slime moulds	–	Form plasmodium under suitable conditions
(3)	Dinoflagellates	–	Have stiff cellulose plates on the cell wall
(4)	Chrysophytes	–	Have two flagella of unequal size

95. The common vegetative reproduction in algae is by
(1) Zoospores
(2) Fragmentation
(3) Conidia
(4) Heterocysts

96. Stems are different from roots as the former
(1) Bear nodes and internodes
(2) Have well developed vascular system
(3) Never have unicellular epidermal appendages
(4) Are always aerial in nature

97. In which of the given pteridophytes, development of zygote for variable period occurs on the parent sporophyte?
(1) *Dryopteris* (2) *Equisetum*
(3) *Selaginella* (4) *Adiantum*

98. Identify the aestivation of petals in the flowers having keel as a characteristic feature.


(1) (2) (3) (4)

99. In the fungi such as *Ustilago*, karyogamy and meiosis take place in
(1) Basidium and basidiospore respectively
(2) Ascus
(3) Basidium
(4) Oogonium

100. At maturity, which elements of phloem lack nucleus?
(1) Only sieve tube elements
(2) Companion cells
(3) Both sieve tube elements and phloem fibres
(4) Only phloem fibres

101. The cell theory that was formulated by Schleiden and Schwann did **not** explain that
(1) Animal cells have a thin outer layer
(2) How were new cells formed
(3) By which structures the plants are composed
(4) What is the unique character of plant cells

102. Which of the following events does **not** occur during S phase of cell cycle?
(1) ATP synthesis
(2) DNA replication
(3) Centriole duplication
(4) Condensation of chromatin

103. Match the following columns and select the **correct** option.

	Column-I		**Column-II**
a.	Smooth endoplasmic reticulum	(i)	Has enzyme that digests nucleic acids
b.	Lysosome	(ii)	Major site for synthesis of lipid
c.	Contractile vacuole	(iii)	Important for osmoregulation
d.	Golgi apparatus	(iv)	Important site of glycoproteins formation

(1) a(ii), b(i), c(iv), d(iii)
(2) a(i), b(ii), c(iv), d(iii)
(3) a(ii), b(i), c(iii), d(iv)
(4) a(iii), b(i), c(ii), d(iv)

104. Chiasmata become visible during which of the following stages of prophase I?
(1) Zygotene (2) Pachytene
(3) Diakinesis (4) Diplotene

105. Turgor pressure of a plant cell will be highest if this cell is placed in

(1) Isotonic solution (2) Pure water

(3) Hypertonic solution (4) Hypotonic solution

106. Select the **correct** match from the following

(1)	Potassium	–	A component of middle lamella in plants
(2)	Calcium	–	Used in the synthesis of cell wall in plants
(3)	Iron	–	Main constituent of coenzyme A
(4)	Boron	–	Needed in the synthesis of auxin

107. Which of the following phenomenon may **not** occur in herbaceous plants if their root pressure is very low?

(1) Guttation

(2) Transpiration

(3) Absorption of minerals

(4) Transport of sugar

108. Free living nitrogen fixing autotrophic microorganisms is

(1) *Nostoc* (2) *Azotobacter*

(3) *Rhizobium* (4) *Beijerinckia*

109. Who first demonstrated that photosynthesis is essentially a light-dependent reaction in which hydrogen reduces carbon dioxide to carbohydrates?

(1) T.W. Engelmann

(2) Julius von Sachs

(3) Cornelius van Niel

(4) Priestley

110. During which of the following conversion steps of respiration, substrate level phosphorylation does **not** occur?

(1) Succinyl CoA → Succinic acid

(2) Glyceraldehyde-3-phosphate → Triose bisphosphate

(3) 1, 2- bisphosphoglyceric acid → 3-phosphoglyceric acid

(4) Phosphoenolpyruvate → Pyruvic acid

111. Select the **incorrect** statement regarding Calvin cycle.

(1) Carboxylation of RuBP results in the formation of 3-carbon molecules

(2) Reduction of CO_2 molecules requires NADPH

(3) The enzyme required during carboxylation step also has oxygenation activity

(4) Regeneration step does not require any ATP molecule

112. In ETS of mitochondria, complex III receives electrons through ubiquinone from

(1) Complex I only

(2) Both complex I and complex VI

(3) Both complex I and complex II

(4) Complex II only

113. All of the following are vegetative propagules in plants, **except**

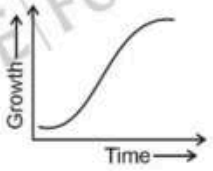
(1) Runner

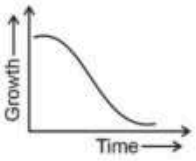
(2) Tuber

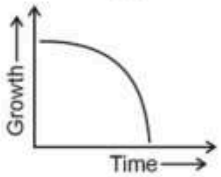
(3) Rhizome

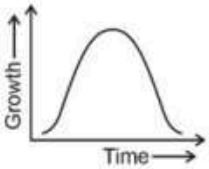
(4) Trichome

114. The growth curve which is characteristic of living organisms growing in a natural environment can be represented as

(1) 

(2) 

(3) 

(4) 

115. Read the following statements and choose the option which is **true** for them

Statement 1 : A majority of plant groups show internal fertilisation.

Statement 2 : The first step of embryogenesis is meiotic cell division.

(1) Both the statements are correct

(2) Both the statements are incorrect

(3) Only statement-1 is correct

(4) Only statement-2 is correct

Akash Medical/IIT-JEE/Foundations

116. "Influencing horizontal growth of seedlings, swelling of the axis and apical hook formation in dicot seedlings."

The above statement is true for

(1) Auxin (2) Ethylene

(3) Cytokinin (4) Gibberellin

117. In an anther, meiotic division occurs in the cells of

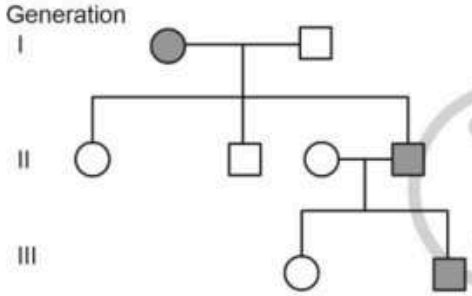
(1) Endothecium

(2) Tapetum

(3) Sporogenous tissue

(4) Middle layer

118. Consider the following pedigree chart and select the option for the set of conclusions given below which can be **true** for the particular character



(i) The trait is Y-linked

(ii) This trait can be autosomal recessive

(iii) This trait can be autosomal dominant

(iv) The male individual in generation I cannot be heterozygous for that character.

(1) (ii) and (iii) only

(2) All (i), (ii), (iii) and (iv)

(3) (i) and (iii) only

(4) (ii) and (iv) only

119. Which of the following **cannot** occur in monoecious species of flowering plants?

(1) Geitonogamy

(2) Autogamy

(3) Xenogamy

(4) Cross pollination

120. In Mendelian dihybrid cross for the colour and shape of pea seeds, the probability of plants in F_2 generation having only one dominant allele for these characters is

(1) 0.15 (2) 0.5

(3) 0.75 (4) 0.25

121. Select the **correct** statement regarding the genes responsible for the morphological characters in *Drosophila*.

(1) Genes for body colour and eye colour are located on both X and Y chromosomes

(2) Distance between the genes responsible for body colour and eye colour is more than that of eye colour and wing size

(3) Genes for body colour and wing size are not on the same chromosome

(4) Genes for body colour, eye colour and wing size are linked genes

122. As compared to DNA, in RNA every nucleotide residue has an additional (i) group present at (ii) in the ribose.

Select the **correct** option to fill in the blanks (i) and (ii).

(1) (i) –COOH, (ii) 2′ position

(2) (i) –OH, (ii) 5′ position

(3) (i) –OH, (ii) 2′ position

(4) (i) –CHO, (ii) 5′ position

123. Which step(s) is/are done before cross hybridisation for developing new varieties of crops?

(a) Testing of superior recombinants.

(b) Testing of new cultivars.

(c) Evaluation of parents.

(d) Collection of variability.

(1) (b) and (c) (2) (a) and (c)

(3) (c) and (d) (4) (a) and (b)

124. The joining of 3′ end of one DNA fragment with 5′ end of other DNA fragment is catalysed by

(1) DNA ligase

(2) DNA helicase

(3) RNA primase

(4) DNA polymerase

125. A variety of chilli which is resistant to mosaic virus is

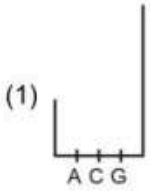
(1) Pusa Komal

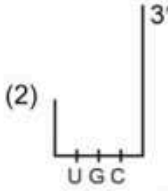
(2) Pusa Sadabahar

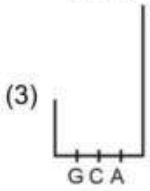
(3) Pusa Shubhra

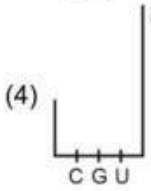
(4) Pusa Swarnim

126. If the coding strand of a DNA has codon 5′ ACG 3′, then which of the following tRNA will carry amino acid for this section of code on mRNA transcribed from this DNA?

(1) 

(2) 

(3) 

(4) 

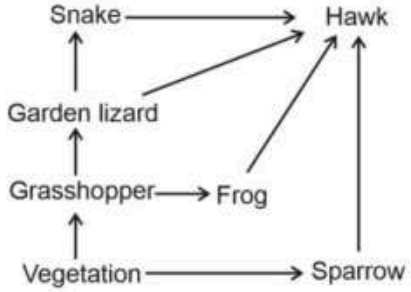
127. The microorganism that is used for making both bread and beer is
(1) *Streptococcus cremoris*
(2) *Saccharomyces cerevisiae*
(3) *Aspergillus niger*
(4) *Streptomyces rimosus*

128. Most of the mammals and birds
(1) Are eurythermal
(2) Cannot regulate their body temperature
(3) Are restricted to narrow range of temperature
(4) Are conformers

129. During sewage treatment, a small part of activated sludge is pumped back into the aeration tank because
(1) A lot of anaerobic bacteria that digest organic material present in sewage water
(2) Microbes present in sludge release toxins that kill harmful bacteria present in sewage water
(3) The sludge has microbes which act as inoculum
(4) The bacteria present in sludge check the growth of putrefying fungi present in sewage water

130. An age pyramid of a population is said to be declining if
(1) The reproductive population is approximately equal to pre-reproductive population
(2) There is fewer individuals in pre-reproductive group as compared to reproductive group
(3) The post-reproductive population is less than the reproductive population
(4) There is more individuals in pre-reproductive group as compared to reproductive group

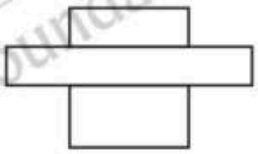
131. In the following food chain, if 100 kilojoule energy is available in the form of food at producer level, then thermodynamically which prey is more efficient for hawk?



(1) Snake (2) Frog
(3) Garden lizard (4) Sparrow

132. Select the statement(s) which is **not true** for a stable community in an ecosystem.
(1) It should not show too much variation in productivity from year to year
(2) It should show highest speed of ecological succession
(3) It must be resistant to natural disturbance
(4) It must be resistant to invasions of alien species

133. A pyramid of number is shown below :



Which of the following food chain corresponds to this pyramid?
(1) Tree → Herbivorous birds → Hawk
(2) Phytoplankton → Zooplankton → Fish
(3) Tree → Herbivorous birds → Parasites
(4) Grass → Grasshopper → Frog

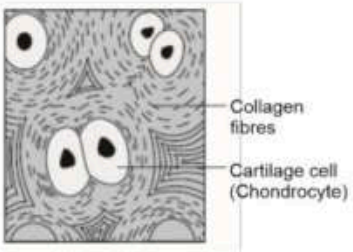
134. Which of the following pollutants is removed from the polluted air by using scrubber?
(1) CO_2 (2) CO
(3) SO_2 (4) H_2S

135. Montreal protocol was signed at Montreal in 1987 to control the
(1) Rate of deforestation
(2) Emission of ozone depleting substance
(3) Emission of greenhouse gases
(4) Production of radioactive chemicals

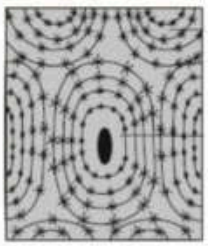
ZOOLOGY

136. Secretion and absorption are main function of a epithelium which is present in
(1) Walls of blood vessels
(2) Tubular parts of nephron
(3) Inner surface of bronchioles
(4) Dry surface of skin

137. Identify the tissues given below in the diagrams and select a character which is shared by both a and b.



(a)



(b)

(1) Fibreless matrix
(2) Presence of lacunae in matrix
(3) Site of erythropoesis in adults
(4) Presence of lamellae in matrix

138. Select the **mismatch** w.r.t. secondary metabolities.
(1) Lectins – Anthocyanins
(2) Alkaloids – Codeine
(3) Drugs – Curcumin
(4) Toxins – Ricin

139. Inadequate secretion of which of the following digestive juice hampers the digestion of disaccharides in small intestine?
(1) Gastric juice (2) Bile juice
(3) Saliva (4) Intestinal juice

140. Among the following five statements (A-E) which option contains all the three **correct** statements regarding marasmus?
A. It is found in infants less than one year in age.
B. It is produced by simultaneous deficiency of proteins and calories.
C. It often happens if the mother has second pregnancy when the older infant is still too young.
D. Extensive oedema and swelling of body parts are seen in it.
E. Characterised by moon like face and pot belly.
(1) A, B, C (2) B, C, D
(3) A, C, D (4) D, C, E

141. Select the **correct** matching of the structure, its location and function from the following given options.

	Structure	Location	Function
(1)	Cerebellum	Forebrain	Temperature regulation
(2)	Crista ampullaris	Membranous labyrinth of inner ear	Maintains the dynamic balance of the body
(3)	Photoreceptor cells	Blind spot	Help in image formation of object on retina
(4)	Limbic system	Midbrain	Contains centre for urge of eating and drinking

142. Read the following statements w.r.t. knee jerk reflex and choose the **incorrect** from them.
(1) There is only one synapse in the reflex pathway
(2) Afferent neuron transmits the impulse into the CNS *via* dorsal nerve root
(3) Efferent neuron synapse with afferent neuron in white matter of spinal cord
(4) Efferent neuron ends at motor end plate of effector

143. Which of the following part of ear is filled with endolymph?
(1) Scala vestibuli
(2) Scala tympani
(3) Vestibular apparatus
(4) Eustachian tube

144. Identify the structures which are **correctly** matched with their correct description.

	Structure	Description
(1)	Scapula and clavicle	Present in each half of pelvic girdle
(2)	Atlas and axis	Cervical vertebrae
(3)	Ilium and ischium	Included in axial skeleton
(4)	Maxilla and zygomatic bones	Form cranium with other cranial bones

145. How many structures/zones from given below in box are present in A bands of myofibrils?

M line, Z line, Actin filament, Myosin filament, H zone

(1) Two (2) Three
(3) Four (4) Five

146. All of the following can lead to osteoporosis in humans **except**
(1) Over secretion of parathormone from parathyroid gland
(2) Deficiency of calcium and vitamin D
(3) Decrease in blood level of sex hormone estrogen in female
(4) High level of Ca^{+2} in body fluids

147. Identify the name of a gland which secretes steroid hormones that stimulate gluconeogenesis, proteolysis and erythropoiesis
(1) Pancreas (2) Kidney
(3) Thyroid (4) Adrenal

148. Read the following statements and choose the **correct** answer.
A. A hormone which stimulates the secretion of hydrochloric acid and pepsinogen from gastric gland is secreted from endocrine cells, present in gastro-intestinal tract
B. Androgens act on the central neural system and influence the male sexual behaviour
(1) Both statements are correct
(2) Both statements are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

149. Match the following hormones (given in column I) with their respective functions (given in column II).

	Column-I		Column-II
a.	Thyroxine	(i)	Stimulates the growth and development of the ovarian follicles in ovaries of females
b.	FSH	(ii)	Regulates the metabolism of carbohydrates, proteins and fats
c.	Thymosins	(iii)	Differentiation of T-lymphocytes
d.	Catecholamines	(iv)	Stimulate the pupillary dilation

Choose the option with all **correct** answer.
(1) a(ii), b(iii), c(i), d(iv)
(2) a(i), b(ii), c(iii), d(iv)
(3) a(ii), b(i), c(iii), d(iv)
(4) a(i), b(iii), c(ii), d(iv)

150. Which of the following statements is **incorrect** w.r.t. human kidney?
(1) Columns of Bertini divided the renal medulla into medullary pyramids
(2) PCT and DCT of all nephrons never extend into the medullary pyramids
(3) The DCTs of many nephrons open into a straight tube called collecting duct, many of which converge and open into the renal pelvis through medullary pyramids in the calyces
(4) A U-shaped vasa recta always runs parallel to the Henle's loop of all nephrons

151. On an average, A ml of blood is filtered by the kidneys per minute which constitute about B of cardiac out put.

Choose the option which represents A and B **correctly**.

	A	**B**
(1)	1100-1200	1/5
(2)	650-700	1/4
(3)	1200-1500	1/3
(4)	1500-2000	1/4

152. During atrial diastole,

(1) Ventricles receive the blood from atria for about 0.4 seconds

(2) Ventricular systole coincides for about 0.4 seconds

(3) The heart does not pump the blood into aorta and pulmonary artery

(4) The atria receive the blood from body and lungs and AV valves remain open for only about 0.3 seconds

153. In ABO blood group system, the person with blood group B can receive the blood from

(1) 'A' blood group person which contains anti B antibodies in their plasma

(2) 'B' blood group person because it does not contain any antibodies in their plasma

(3) 'O' blood group person which does not possess any antigen on surface of their RBCs

(4) 'AB' blood group person which contains both A and B antigen on surface of their RBCs

154. Select the function of most abundant leucocytes among the following.

(1) Destroy foreign organisms by phagocytosis

(2) Secrete histamines and serotonin

(3) Resist infection and combat allergic reaction

(4) Provide cell mediated immunity

155. The blood vessel which carry deoxygenated blood away from heart, begins from

(1) Left ventricle

(2) Aorta

(3) Pulmonary vein

(4) Right ventricle

156. Choose the **incorrect** match w.r.t. cockroach.

(1)	Arthrodial membrane	–	Joins the sclerites
(2)	Mushroom-shaped gland	–	Present in the 6th-7th abdominal segments of male
(3)	Tegmina	–	Develops after the last moulting in adult cockroach
(4)	Alary muscles	–	12 pairs, present in ventral side of the body

157. Which one of the following categories of animals is **correctly** described?

(1) Reptiles possess external ear openings

(2) Bony fishes lack the capacity to regulate their body temperature

(3) Cyclostomes are fresh water fishes but migrate for spawning to marine water

(4) Annelids are mostly dioecious and exhibit internal fertilisation

158. Match the name of the animal (Column-I), with one characteristic (Column-II), and the phylum/ class (Column-III) to which it belongs and select the **correct** answer.

	Column-I	**Column-II**	**Column-III**
(1)	*Antedon*	Chitinous exoskeleton outside the body	Arthropoda
(2)	*Aedes*	Circulatory system is of closed type	Annelida
(3)	*Carcharodon*	Gill slits are separate and without operculum	Chordata
(4)	*Myxine*	Air bladder regulates buoyancy	Osteichthyes

159. Consider the following features

(a) Internal fertilisation (b) Metamerism

(c) Oviparity

Select the **correct** option of animals which possess all the above features

(1) *Nereis, Hyla, Bombyx*

(2) *Pheretima, Clarias, Panthera*

(3) *Neophron, Apis, Chelone*

(4) *Locusta, Planaria, Hyla*

160. If a circular dsDNA possess three cloning sites for *EcoRI* and was subjected to digestion with same restriction enzyme, after digestion, mixture contained

(1) 3 fragments (2) 4 fragments

(3) 5 fragments (4) 6 fragments

161. Choose a selectable marker which is present in both pUC8 and pBR322

(1) *Lac* Z coding for β galactosidase

(2) Ampicillin resistance gene

(3) Tetracycline resistance gene

(4) Kanamycin resistance gene

162. Some of the important factors, listed below are present in both simple and sparged stirred tank biroreactor **except**

(1) Maintenance of aseptic condition

(2) Anti-foaming agents to reduce the foaming caused by aeration

(3) Provision for control of temperature and pH

(4) Sparger to increased surface area of oxygen transfer

163. Bt crops are transgenic plants which are useful to

(1) Make the crops less tolerant to abiotic stresses

(2) Reduce reliance on chemical pesticides

(3) Increasing post harvest losses

(4) Reduce nutritional value of food

164. Select the **odd** one w.r.t. recombinant therapeutics which have been approved globally and presently marketed in India

(1) FSH

(2) Streptokinase

(3) Tissue plasminogen activator

(4) Hepatitis vaccine

165. Arrange the below given steps of human insulin formation in a proper sequence and choose the **correct** option.

(a) Separation and extraction of A and B chains of insulin from bacteria

(b) Prepared DNA sequences corresponding to A and B chains were introduced in plasmid pBR322

(c) Introduction of rDNA into host *E. coli* for multiplication

(d) Artificial synthesis of nucleotide sequence corresponding to A and B chains of human insulin.

(e) Formation of human insulin by creating disulphide bonds between A and B chains

(1) (b) → (c) → (d) → (a) → (e)

(2) (a) → (b) → (c) → (d) → (e)

(3) (d) → (b) → (c) → (a) → (e)

(4) (d) → (b) → (c) → (e) → (a)

166. Which one is matched **incorrectly**?

(1)	Fisheries	–	Catching, processing and selling of fish or other aquatic animals
(2)	Aquaculture	–	Involves production of only useful aquatic animals
(3)	Interspecific hybridisation	–	Mating between the parents of two different species
(4)	Poultry	–	Rearing and breading of fowls for eggs and food

167. Fowl cholera is a

(1) Bacterial disease of cattles

(2) Bacterial disease of birds

(3) Viral disease of birds

(4) Viral disease of cattles

168. Which of the following is an example of cross breed?

(1) Mule

(2) Gir

(3) *Hisardale*

(4) Jersey

169. Select the **incorrect** property/character for antibodies
(1) Glycoproteins
(2) Immunoglobulins
(3) Produce in response to antigenic stimulation
(4) Produced by natural killer cells

170. What is **not** true about B-lymphocytes in human?
(1) They are involved in primary as well as secondary immune response
(2) They require T-lymphocytes to exhibit humoral immune response
(3) Thymus gland provides microenvironment for their development and maturation
(4) They are not responsible majorly for graft rejection

171. Read the following statements and select the **correct** answer.
A. Drugs for HIV inhibits HIV replication and slow the progression of AIDS.
B. Enzyme reverse transcriptase reverse transcribes HIV RNA genome to form viral DNA.
(1) Both statements are correct
(2) Both statements are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

172. Animal which probably evolved from thecodonts is
(1) Mammals (2) Dinosaurs
(3) Tuataras (4) Lizards

173. Choose the **correct** set of plants which probably evolved from seed ferns
(1) Cycads, dicotyledons
(2) Gnetales, conifers
(3) Ferns, Ginkgos
(4) Zosterophyllum, Sphenopsids

174. In Australia, sugar glider, koala, wombat have evolved to share many similar characteristics. This type of evolution may be referred to as
(1) Adaptive convergence
(2) Convergent evolution
(3) Adaptive radiation
(4) Co-evolution

175. Select the option which **correctly** represents the chromosome numbers in gamete of dog and cat respectively
(1) 46, 12 (2) 12, 42
(3) 39, 19 (4) 21, 19

176. The offsprings are extremely vunerable to predators in
(1) Reptiles and birds
(2) Bony fishes and frogs
(3) Sponges and terrestrial arthropods
(4) All sexually reproducing organisms

177. Process capacitation occurs in
(1) Sperms in female genital tract before acrosomal reaction
(2) Ovum in fallopian tube before fertilisation
(3) Primary oocyte in ovary during oogenesis
(4) Primary spermatocytes in seminiferous tubules during spermatogenesis

178. At which time in a normal menstrual cycle does rapid secretion of estrogen occurs?
(1) In the middle of follicular phase
(2) At the end of proliferative phase
(3) At the end of luteal phase
(4) In the middle of menstrual phase

179. In humans, the blastocyst is a
(1) Haploid structure which is implanted in uterine endometrium
(2) Diploid structure which are formed at the end of cleavage
(3) Embryonic structure which is characterised by the presence of extraembryonic foetal membrane
(4) Structure which exhibits gastrulation to form only two germ layers

180. During oogenesis, the second polar body is formed
(1) In primary follicle
(2) By secondary oocyte in fallopian tube
(3) At completion of meiosis 1st
(4) In ovum, within ovary

❑ ❑ ❑